

DIALLO BOYE

Quantitative Researcher | Systematic Strategies

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PROFILE

Quantitative researcher with 6+ years at CERN and Brookhaven National Laboratory, specialising in statistical inference, signal extraction from high-dimensional noisy datasets, and end-to-end model development under strict uncertainty quantification constraints. Experienced in leading international research teams, delivering peer-reviewed results, and translating complex quantitative findings into actionable insights. Transitioning into quantitative finance.

CORE COMPETENCIES

Statistical Inference Hypothesis testing, Bayesian & frequentist methods, likelihood-free inference, limit setting	Signal Extraction & Anomaly Rare signal detection in high-noise environments, multivariate classifiers, background modeling	Uncertainty Quantification Calibrated confidence intervals, systematic uncertainty propagation, sensitivity analysis
Time Series & Stochastic Modeling Non-stationary processes, trend extraction, precision measurement, sub-percent accuracy	Machine Learning & Optimisation Neural networks, gradient-based optimisation, surrogate models, simulation-based inference	Programming Python (NumPy, pandas, scikit-learn, SciPy), C++, SQL, Linux/bash

PROFESSIONAL EXPERIENCE

Goldhaber Distinguished Fellow - Brookhaven National Laboratory / CERN 2023 – 2026

Highly competitive fellowship awarded to fewer than 5 early-career scientists per year worldwide.

- ▶ Led an international quantitative research group of ~20 scientists (Apr 2025 – present): set research agenda, allocated resources across concurrent projects, and oversaw delivery of results — directly analogous to running a systematic research team.
- ▶ Developed and deployed signal extraction models on one of the world's largest datasets, combining multivariate classifiers, likelihood-based inference, and rigorous uncertainty quantification to identify rare events against overwhelming backgrounds.
- ▶ Built automated ML pipelines replacing thousands of hand-tuned individual fits — improving robustness, scalability, and precision of production results delivered to a 3,000+ scientist collaboration.
- ▶ Invited plenary speaker at LHCP 2026 (Paris) — selected to present a cross-experiment review to 500+ scientists; 6 further international conference talks.

Postdoctoral Research Associate - Brookhaven National Laboratory 2020 – 2023

- ▶ Developed time series analysis tools for beam parameter evolution: extracting non-stationary trends, modeling correlations, and delivering sub-percent precision calibration results.
- ▶ Mentored 4 graduate students across 3 continents in quantitative methods, Python, and research methodology.

PhD Researcher - CERN / ATLAS 2016 – 2020

- ▶ Built end-to-end quantitative pipelines: Monte Carlo simulation, multivariate classification, background modeling, statistical hypothesis testing, and limit setting — published in Physics Letters B.

PROJECTS

swiss-power-forecast - Probabilistic Electricity Price Forecasting github.com/dialloboyce/swiss-power-forecast

- ▶ Swiss day-ahead price forecasting (ENTSO-E): LightGBM beats the naive baseline by 29% MAE in walk-forward backtesting; overconfident intervals (56.5% vs 80% coverage) diagnosed and repaired via conformalized quantile regression (79.1%).

market-regime-scan - Multiple Testing in Regime Detection github.com/dialloboyce/market-regime-scan

- ▶ Bump-hunt scan of 26 years of S&P 500 volatility: 27 local “3 σ ” regime changes reduce to 1 under a block-bootstrap null; look-elsewhere effect quantified with a three-null ladder (18 $\rightarrow$ 5 $\rightarrow$ 1 survivors).

shrunk-en-portfolios - Portfolio Optimization & VaR Backtesting github.com/dialloboyce/shrunk-en-portfolios

- ▶ 20-year walk-forward horserace: the max-Sharpe optimizer finishes last on Sharpe (0.30 vs 0.74 for 1/N); Ledoit-Wolf shrinkage from the paper; regulatory VaR backtests (Kupiec, Christoffersen, Basel) — EWMA + Student-t restores calibration.

EDUCATION

PhD, Particle Physics - CERN / ATLAS / University of South Africa 2020

MSc, Physics of Energy & Energy Transition - Universite Paul Sabatier, Toulouse 2016

MSc, Subatomic Physics & Astroparticle - Universite de Strasbourg 2014

LANGUAGES French Native English Fluent